

News-Driven Price Impulses and Bubble Amplification Effects in AI Technology Stocks

Population

Daily stock prices and Bloomberg financial news articles for five US AI technology stocks (NVDA, META, MSFT, AMZN, GOOGL) from 2021 to 2025

Primary Objective

To see whether AI-related news sentiment can predict the next-day return and price direction of AI technology stocks, and to compare how well different models do this

Secondary Objectives

To look at whether the US monetary policy environment changes how much news affects stock prices

To check whether AI stocks show more bubble-like price behaviour than the broader market over this period

To compare FinBERT sentiment scoring with a traditional word-based approach and see which works better

Data Collection

Stock price data comes from Yahoo Finance and covers daily returns and closing prices for all five stocks from 2021 to 2025

News articles are collected from Bloomberg by searching each stock name with relevant keywords, and the date, title, and summary of each article are saved in Excel files

US Federal Funds Rate and FOMC meeting decisions from 2021 to 2025 are used to define three monetary policy phases

Variables Under Consideration

Next-day stock return — what we are trying to predict, as a continuous value

Next-day price direction, up or down — derived from the daily return, used for directional accuracy evaluation

Article title and summary text — the raw news input used to compute sentiment

Daily article count per stock — used to decide whether a day has enough news coverage to be included

Federal Funds Rate — used to capture the monetary policy environment

Monetary policy phase — three periods defined by FOMC decisions, used to compare model behaviour across regimes

Missing Data Procedures

Days with fewer than 2 articles or where the model has low confidence in the sentiment are left out of the analysis, and no imputation is done

Missing macroeconomic values on non-trading days are filled in using the most recent available figure

Numerical and Graphical Summaries

Summary statistics for daily returns of each stock broken down by monetary policy phase

Time series plots of daily returns with the three policy phase boundaries marked

Distribution of sentiment scores by news category

Bubble detection plots showing periods of explosive price growth for each stock compared to the S&P 500

Predicted return intervals plotted against actual next-day returns

Model accuracy comparison across stocks and monetary policy phases

Models to be Fitted

A news sentiment model is used to predict next-day returns, where sentiment scores from FinBERT are fed into a time series framework; monetary policy phase is included to test whether it changes how strongly news moves prices

A deep learning model taking historical prices and sentiment as inputs is fitted as an alternative and compared against the sentiment model

A bubble detection method is applied to the price series of each stock and the S&P 500 to find periods of explosive growth and compare them across monetary policy phases

FinBERT and a word-based sentiment method are compared on the same articles

Models are judged on three things: whether directional accuracy is above 50%, whether the prediction interval excludes zero, and whether actual returns fall inside the predicted interval